



Data Analyst Internship Home Assignment: User Segmentation & Business Insights

Overview & Approach

Business Objective: The company aims to gain deeper insights into user behaviour on its blog platform in order to enhance user engagement and boost retention.

Dataset Summary: This dataset is a synthetic simulation of 100,000 blog users, capturing key behavioural metrics such as:

- ▶ Engagement (views, comments, support tickets volume, sessions)
- ▶ Content Creation (posts published, draft count, post length, images per post)
- ▶ Subscription Status (Free vs. Paid)
- ▶ Platform Usage (mobile vs. desktop, SEO setup, feature adoption)

Objective of User Segmentation:

The goal of segmentation was to better understand data driven insights of user behaviour patterns on a blog platform to improve user engagement and retention. By identifying distinct user groups, we can tailor strategies to boost engagement, optimize content reach, and enhance the user experience.

Segmentation Methodology

Approach Used: K-Means Clustering in BigQuery ML

I applied K-Means clustering in Big Query ML, an unsupervised machine learning technique, to categorize users into **four** meaningful segments based on their activity levels. The clustering was performed using the following behavioural metrics:

- ▶ **Posts Published** – Total number of blog posts published by the user.
- ▶ **Total Views** – Total number of post views across all posts.
- ▶ **Total Comments** – Total number of post comments received.
- ▶ **Total Sessions Last 30 Days**– Number of sessions the user had in the last 30 days.

These features were chosen because they reflect both **content creation (posts)** and **user engagement (views, comments, and sessions)**, allowing for a well-rounded segmentation.

Segmentation Methodology

Why This Approach?

- ▶ **Data-Driven & Scalable:** K-Means clustering allowed for an objective, data-driven segmentation rather than random thresholds.
- ▶ **Behaviour-Based:** The clusters were built directly from user activity, making it actionable for marketing and retention strategies.
- ▶ **Business Impact:** The identified clusters align with business goals, enabling targeted interventions to improve user engagement and experience.

Segmentation Methodology

Data Preparation

Before running the K-Means model, the following data preparation steps were taken:

1. **Data Cleaning & Filtering:** Ensured no duplicate or invalid data was present in the dataset.
2. **Handling Missing Values:** Checked the dataset for missing data.

Segment Profiles & Findings

Based on clustering results, I identified the following **four** user segments:

1. Unoptimized Users
2. Engaged Users
3. Casual Users
4. Key Contributors

Segment	# users	# AVG Posts	# AVG Views	# AVG Comments	# AVG Sessions	% SEO Setup (Yes)	% SEO Setup (No)	% Subscription (PAID)	% Subscription (FREE)
Unoptimised Users	4,769	2,745	2	0	11	0.00%	100.00%	0.00%	100.00%
Engaged Users	24,410	270	14,078	144	70	88.75%	11.25%	79.86%	20.14%
Casual Users	65,248	13	34,499	7	11	40.25%	59.75%	41.29%	58.71%
Key Contributors	5,573	9	3,543,093	275	23	99.98%	0.02%	50.12%	49.88%

Segment Profiles & Findings

Unoptimized Users

Segment	# users	# AVG Posts	# AVG Views	# AVG Comments	# AVG Sessions	% SEO Setup (Yes)	% SEO Setup (No)	% Subscription (PAID)	% Subscription (FREE)
Unoptimised Users	4,769	2,745	2	0	11	0.00%	100.00%	0.00%	100.00%

Definition: Users who create a high number of posts but receive little to no engagement (low views, comments, and sessions).

Characteristics:

- ▶ **High content production** but **low audience reach**.
- ▶ **All users have a FREE subscription** and **haven't completed SEO setup**.

Segment Profiles & Findings

Engaged Users

Segment	# users	# AVG Posts	# AVG Views	# AVG Comments	# AVG Sessions	% SEO Setup (Yes)	% SEO Setup (No)	% Subscription (PAID)	% Subscription (FREE)
Casual Users	65,248	13	34,499	7	11	40.25%	59.75%	41.29%	58.71%

Definition: Users who actively create content and receive strong engagement (high views, comments, and sessions).

Characteristics:

- ▶ **Moderate content creation** and **strong engagement**.
- ▶ Users are actively engaging with the platform and growing their audience.
- ▶ The **majority are leveraging SEO practices** and **have PAID subscriptions**.

Segment Profiles & Findings

Casual Users

Segment	# users	# AVG Posts	# AVG Views	# AVG Comments	# AVG Sessions	% SEO Setup (Yes)	% SEO Setup (No)	% Subscription (PAID)	% Subscription (FREE)
Engaged Users	24,410	270	14,078	144	70	88.75%	11.25%	79.86%	20.14%

Definition: Users who occasionally publish content and receive high views but very few comments, with limited sessions.

Characteristics:

- Low number of posts.
- Few comments per post despite high audience reach.
- Might be interested, but not fully committed to maintaining an active blog.
- More than half users are NOT leveraging SEO practices and have FREE subscription.

Segment Profiles & Findings

Key Contributors

Segment	# users	# AVG Posts	# AVG Views	# AVG Comments	# AVG Sessions	% SEO Setup (Yes)	% SEO Setup (No)	% Subscription (PAID)	% Subscription (FREE)
Unoptimised Users	4,769	2,745	2	0	11	0.00%	100.00%	0.00%	100.00%

Definition: Possibly highly influential users (e.g., popular bloggers, viral content creators) who drive the majority of traffic on the platform.

Characteristics:

- **Very low content production** but **very high engagement**.
- Core contributors to platform growth and influence.
- **The majority are leveraging SEO practices while the subscription distribution is nearly even.**

Key Insights & Recommendations

Summary of Key Takeaways:

The segmentation analysis revealed distinct user behaviours, highlighting opportunities to improve platform engagement and retention across segments:

- **Unoptimized Users:** High content production but low engagement due to limited visibility, linked to free subscriptions and lack of SEO setup. Educating them on SEO and visibility tools could help.
- **Engaged Users:** Strong content engagement and audience growth, benefiting from SEO and paid subscriptions, making them an ideal balance of creators and interactors.
- **Casual Users:** Occasional posters with some reach but minimal engagement, showing untapped potential due to infrequent platform use. Over half don't use SEO and rely on free subscriptions, missing growth opportunities.
- **Key Contributors:** Low content production but high engagement, driving most platform traffic. Retaining them with exclusive perks or monetization could enhance loyalty.

Key Insights & Recommendations

Notable Patterns

- Posting more does not necessarily lead to more engagement. **SEO setup** and **content visibility** are likely key factors.
- **Consistency** in content creation **correlates with increased engagement**.
- **Quality** and **influence matter more than quantity** for engagement.

Key Insights & Recommendations

Actionable Ideas to Improve Engagement & Product Value

1. Educational Campaign for Unoptimized Users

- ❑ Introduce SEO setup prompts in the platform.
- ❑ Provide free SEO workshops or tutorials to help users optimize their content.
- ❑ Add automated recommendations for improving post visibility (e.g., "Your post could reach x% more users with an SEO title").

Expected Impact: Increased discoverability for users who haven't completed SEO setup → higher views and engagement.

2. Re-engagement Strategies for Casual Users

- ❑ Introduce notifications or reminders to prompt them to post new content or engage with their audience.
- ❑ Use personalized email nudges (e.g., "Your last post got 500 views! Why not share another update?").
- ❑ Gamify the experience with badges for consistent posting and interacting.

Expected Impact: Casual Users move toward the Engaged Users category, increasing content volume and engagement.

Key Insights & Recommendations

Actionable Ideas to Improve Engagement & Product Value

3. Exclusive Perks for Key Contributors

- ❑ Offer exclusive benefits like premium features, early access to new tools, or monetization options to show appreciation and encourage loyalty.
- ❑ Recognize their impact by highlighting them in a “Top Creators” section or giving them special badges.
- ❑ Leverage their influence by inviting them to become Brand Ambassadors, representing the platform through campaigns, referral programs, and special events. Their role as ambassadors can drive platform growth and strengthen their connection to the community.

Expected Impact: Offering exclusive perks, recognition, and ambassador roles to Key Contributors is expected to enhance retention, boost content quality, drive organic platform growth, and strengthen community engagement.

Conclusion

This analysis shows that content production and engagement don't always align—some users post frequently with low visibility, while others create little but drive high engagement.

The four segments reveal distinct behaviours:

- **Unoptimized Users** post often but lack SEO, limiting reach.
- **Engaged Users** balance content and engagement, setting a success model.
- **Casual Users** have growth potential but post infrequently with low engagement.
- **Key Contributors** drive the most engagement despite minimal content.

These insights help tailor strategies to boost visibility, participation, and community interaction.

Conclusion

Next Steps for Further Analysis:

- **Content Quality Analysis** – What types of content drive the most engagement? Are there themes or formats that outperform others?
- **User Retention Trends** – How do these segments evolve over time? Do Casual Users become Engaged Users, or do they drop off?
- **A/B Testing of Engagement Strategies** – Experiment with SEO education, personalized nudges, and community features to measure impact.

Appendix

SQL Query for K-Means Clustering

```
SELECT *  
  
FROM  
  
ML.PREDICT( MODEL `my-project-454915.blog_platform.user_clusters`,  
  
(SELECT *  
  
FROM blog_platform.users)  
  
)
```